

Stack:

Activity-1

29-7-28
19252515
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used

* push (54) $\rightarrow$ (54)

* push (52) $\rightarrow$ (54, 52)

* pop() $\rightarrow$ removes 52

* push (55) $\rightarrow$ (54, 55)

* push (62) $\rightarrow$ (54, 55, 62)

* $s = \text{pop}() = 62$

Queue:-

* enqueue (21) $\rightarrow$ (21)

* enqueue (24) $\rightarrow$ (21, 24)

* dequeue() $\rightarrow$ removes 21

* enqueue (28) $\rightarrow$ (24, 28)

* enqueue (32) $\rightarrow$ (24, 28, 32)

* $q = \text{dequeue}() = 24$

Answer $s + q = 62 + 24 = 86$

2. postfix Expression:

Expression: $105 + 60 / 6 * 8 -$

Evaluation:

* $10 + 5 = 15$

* $60 \div 6 = 10$

* $15 \times 10 = 150$

* $150 - 8 = 142$

Answer = 142

3. postfix expression

expn: $8\ 2\ 3\ ^\wedge / 2\ 3\ ^\wedge + 5\ 1\ ^\wedge$

After the first $^\wedge$ is evaluated $\left[\frac{2^3}{8} \right] \left[\frac{8}{1} \right] \left[\frac{3}{1} \right] \left[\frac{6}{1} \right]$

$^\wedge$ stack becomes $[1, 6]$ 6, 1

Top two elements - 6 and 1 ($\text{Top} = 6$)

Ans = 6, 1

4. Infix to postfix

Expression: $a + b \times c - d \wedge e \wedge f$

Operator precedence

1. $\wedge$ (highest, right associative)

2. $\times$

3. $+$, $-$

Step by step

1. $b \times c \rightarrow bc^\times$

2. $a + (bc^\times) \rightarrow abc^\times +$

3. $e \wedge f \rightarrow ef^\wedge$

$d \wedge (ef^\wedge) \rightarrow def^\wedge\wedge$ (because $\wedge$ is right associative)

5. $(a + bc) - (d \wedge (e \wedge f))$

Postfix: $abc^\times + def^\wedge\wedge -$

5. Stack operation

operation:

- ~~PUSH(10)~~
- PUSH(20)
- POP $\rightarrow$ ~~20~~
- PUSH(20)
- ~~PUSH(20)~~
- POP $\rightarrow$ 20
- POP $\rightarrow$ 10
- POP $\rightarrow$ 10
- ~~PUSH(20)~~
- POP $\rightarrow$ 20

Sequence of popped values

20, 20, 10, 10, 20

Amal
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